Manuscript submission

Finite Fields and Their Applications 26 October 2026

Dear Editors,

Please consider the manuscript *The Welch function is not third-order sum-free* by Deep Bhattacharjee for publication in *Finite Fields and Their Applications*.

The manuscript proves that the Welch function X^{2^m+3} on $\mathbb{F}_{2^{2m+1}}$ is not third-order sum-free for every $m \geq 3$, settling the conjecture formulated by Hou and Zhao. It gives a two-trace criterion and the exact count $(2^n + 1 - 3K_n)/4$, with K_n a binary Kloosterman sum. The corresponding number of three-dimensional linear zero-sum subspaces is $N_n(2^n - 1)/42$. The proof is written out from the reduction to the final character-sum count and uses the classical Weil bound. The supplied code checks the formulas independently, and the supplied Lean files check specified identities and small cases. The finite-field structure, character sums and relation to APN functions fit the journal's scope.

An earlier version is publicly available as Hexagon 2610.00472v1. The source, verification code and supporting material are archived with the accompanying release:

`https://github.com/creelie/eConjectureWelchSumFree/releases/tag/v1.2.0`

The submission includes a separate declaration describing AI assistance and the scope of the formal verification.

Thank you for considering the manuscript.

Sincerely,

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